MATHEMATICS BY D.K. SINGH

JEE MAINS PRACTICE TEST

Section A: Single Correct Choice Type

Marking Scheme: +4 for a correct answer, –1 for an incorrect answer, and 0 if left unattempted. This section contains **20** multiple-choice questions.

Q.1 The expression $2(1+\cos x)-\sin^2 x$ is the same as:

(A) $1+\cos^2 x$
(B) $(1+\sin x)^2$
(C) $1-\cos^2 x$
(D) $(1+\cos x)^2$

Q.2 The value of $\dfrac{\sin 10^\circ+\sin 20^\circ}{\cos 10^\circ+\cos 20^\circ}$ equals:

(A) $2+\sqrt{3}$
(B) $\sqrt{2}-1$
(C) $2-\sqrt{3}$
(D) $\sqrt{2}+1$

Q.3 Which of the following quantities is irrational?

(A) $\sin\left(\dfrac{11\pi}{12}\right)\sin\left(\dfrac{5\pi}{12}\right)$
(B) $\sin^4\left(\dfrac{\pi}{8}\right)+\cos^4\left(\dfrac{\pi}{8}\right)$
(C) $\left(1+\cos\dfrac{\pi}{9}\right)\left(1+\cos\dfrac{2\pi}{9}\right)\left(1+\cos\dfrac{3\pi}{9}\right)$
(D) $\sin\left(\dfrac{5\pi}{12}\right)\cos\left(\dfrac{5\pi}{12}\right)\sin\left(\dfrac{9\pi}{12}\right)$

Q.4 The solution set of $|x-1|-2|x|=3|x+1|$ contains:

(A) Exactly three elements
(B) Exactly two elements
(C) More than 3 elements
(D) Exactly one element

Q.5 The set of all real numbers x that satisfy $\dfrac{x-2}{x+2}>\dfrac{2x-3}{4x-1}$ is:

(A) $(-\infty,-2)\cup\left(\dfrac{1}{4},\dfrac{3}{2}\right)\cup(2,\infty)$
(B) $\left(-2,\dfrac{1}{4}\right)\cup\left(\dfrac{3}{2},2\right)$

(C) $(-\infty, -2) \cup \left(\frac{1}{4}, 1\right) \cup (4, \infty)$

(D) $\left(-2, \frac{1}{4}\right) \cup (1, 4)$

Q.6 The number of values of x in $[0, 10\pi]$ for which $(\sin x)^{15} - (\cos 2x)^{19} = 2$ is:

(A) 0

(B) 5

(C) 10

(D) None

Q.7 The sum $\sqrt{\frac{5}{4} + \sqrt{\frac{3}{2}}} + \sqrt{\frac{5}{4} - \sqrt{\frac{3}{2}}}$ is equal to:

(A) $\tan \frac{\pi}{3}$

(B) $\cot \frac{\pi}{3}$

(C) $\sec \frac{\pi}{3}$

(D) $\sin \frac{\pi}{3}$

Q.8 Suppose p, q, r and $s \in \mathbb{N}$ satisfy the relation $p + \cfrac{1}{q + \cfrac{1}{r + \cfrac{1}{s}}} = \frac{89}{68}$, then find the value of $(pq + rs)$.

(A) 19

(B) 21

(C) 23

(D) 25

Q.9 Sum of roots of the equation $(x+3)^2 - 4|x+3| + 3 = 0$ is:

(A) 4

(B) 12

(C) -12

(D) -4

Q.10 If $x = \sin\theta|\sin\theta|$, $y = \cos\theta|\cos\theta|$ where $\theta \in \left[\frac{99\pi}{2}, 50\pi\right]$, then:

(A) $x - y = 1$

(B) $x + y = -1$

(C) $x + y = 1$

(D) $y - x = 1$

Q.11 If $\frac{3\pi}{4} < \theta < \pi$, then $\sqrt{\frac{2}{\tan\theta} + \frac{1}{\sin^2\theta}}$ equals:

(A) $1 + \cot\theta$

(B) $\cot\theta - 1$
(C) $1 - \cot\theta$
(D) $-1 - \cot\theta$

Q.12 If $\sin A = \dfrac{3}{\sqrt{13}}$, $\cos B = \dfrac{5}{\sqrt{26}}$ where A is an obtuse angle and B is an acute angle, then $A+B$ equals:

(A) $135°$
(B) $225°$
(C) $145°$
(D) $165°$

Q.13 The product of all values of x satisfying the equation $|x+1| - |x| + 3|x-1||x-2| = x+2$ is:

(A) $\dfrac{5}{3}$
(B) $\dfrac{25}{9}$
(C) $\dfrac{5}{9}$
(D) 5

Q.14 Let $\theta \in \left(\dfrac{\pi}{2}, \pi\right)$, then the expression $\Big|\big||\cos\theta - 3\sin\theta| - 2\sin\theta\big| + \cos\theta - \tan\theta\Big|$ simplifies to:

(A) $\sin\theta - \tan\theta$
(B) $\cos\theta - \tan\theta$
(C) $-\cos\theta - \tan\theta$
(D) $-\dfrac{3}{2}\tan\theta$

Q.15 If $\sin\theta + \csc\theta = 2$, then the value of $\sin^8\theta + \csc^8\theta$ is equal to:

(A) 2
(B) 2^4
(C) 2^8
(D) More than 2^8

Q.16 If θ is an acute angle and $\tan\theta = \dfrac{1}{\sqrt{7}}$, then the value of $\dfrac{\csc^2\theta - \sec^2\theta}{\csc^2\theta + \sec^2\theta}$ is:

(A) $3/4$
(B) $1/2$
(C) 2
(D) $5/4$

Q.17 If $\dfrac{\pi}{2} < \theta < \dfrac{3\pi}{2}$, then $\sqrt{\tan^2\theta - \sin^2\theta}$ is equal to:

(A) $\tan\theta\sin\theta$
(B) $-\tan\theta\sin\theta$
(C) $\tan\theta - \sin\theta$

(D) $\sin\theta - \tan\theta$

Q.18 The minimum value of $f(x) = |x-1| + |x-2| + |x-3|$ is equal to:

(A) 1
(B) 2
(C) 3
(D) 0

Q.19 If the expression $\cos\left(x - \frac{3\pi}{2}\right) + \sin\left(\frac{3\pi}{2} + x\right) + \sin(32\pi + x) - 18\cos(19\pi - x) + \cos(56\pi + x) - 9\sin(x + 17\pi) = A\sin x + B\cos x$, then find the value of $\frac{B}{A}$.

(A) 1
(B) $\frac{1}{2}$
(C) -1
(D) 2

Q.20 The value of $\left|-\cot\frac{\pi}{32} + \tan\frac{\pi}{32} + 2\tan\frac{\pi}{16} + 4\tan\frac{\pi}{8}\right|$ is:

(A) 8
(B) 9
(C) 6
(D) 16

Section B: Numerical Value / Integer Answer Type

Marking Scheme: +4 for a correct answer, 0 for incorrect or unattempted. This section contains **5** questions.

Q.21 Find the value of the expression: $\cot 2^\circ \cot 3^\circ - \cot 3^\circ \cot 5^\circ - \cot 5^\circ \cot 2^\circ$.

Q.22 Find the number of ordered pairs (m, n) where $m, n \in \mathbb{N}$ such that $m^2 = n^2 + 2022$.

Q.23 Find the largest integral value of a for which the system of equations

$$\begin{aligned} x + ay &= 3 \\ ax + 4y &= 6 \end{aligned}$$

satisfies $x > 1$ and $y > 0$.

Q.24 Find the number of integral values of x which simultaneously satisfy the inequalities $|x| + 5 < 7$ and $|x - 3| > 2$.

Q.25 If the solution set of $|x - k| < 2$ is a subset of the solution set of the inequality $\dfrac{2x - 1}{x + 2} < 1$, then find the number of possible integral values of k.